

Q.24 Draw and explain the block diagram of a Pyroelectric Vidicon.

Q.25 Explain in detail the principles and applications of PET imaging.

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4th Semester/ Medical Electronics
Subject : Advance Medical Imaging Technique

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which is a component of a Digital Radiography system?

- a) Collimator b) Digital detector
- c) Motor coil d) Thermistor

Q.2 Which CT detector uses gas for X-ray absorption?

- a) Photomultiplier b) Scintillation
- c) Xenon d) Crystal diode

Q.3 Which radioactive emission has the highest penetration?

- a) Alpha b) Beta
- c) Gamma d) None

Q.4 Which factor describes how much radiation is absorbed by the body?

- a) Reflectivity b) Transmittance
- c) Absorptivity d) Emissivity

Q.5 The full form of PET is:

- a) Photonic Emission Tomography
- b) Positron Emission Tomography
- c) Photon Energy Transmission
- d) Prolonged Exposure Technique

Q.6 Which type of radiation is non-ionizing?

- a) Gamma rays b) X-rays
- c) Infrared rays d) Beta rays

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Expand AMIT.

Q.8 Write one use of slip rings in CT scanners.

Q.9 Mention one component of Thermographic equipment.

Q.10 Define Radioactive decay.

Q.11 Expand CT.

Q.12 Write one risk factor of radiation exposure.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write a short note on the components of a DR system.

Q.14 Explain the role of slip rings in CT scanners.

Q.15 Write a short note on gamma emission.

Q.16 Write a short note on electron capture.

Q.17 Write a short note on reflectivity.

Q.18 Explain the principle of a Pyroelectric Vidicon.

Q.19 Write a short note on positron emission tomography (PET).

Q.20 Explain the maintenance procedure for X-ray machines.

Q.21 Write four advantages of DSA.

Q.22 Briefly explain radiation safety limits.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw the block diagram of Thermographic equipment and explain its working.